

DocIntel — Intelligent Document Extraction, Validation & API Platform

- AI Engineer Internship Case Study
- Four financial document types • OCR • structured extraction • financial validation • persistent dashboard

Problem & Objective

- Financial teams receive native/scanned PDFs and images with varying layouts and OCR quality.
- Goal: convert documents into reliable structured JSON, validate financial relationships, store results and expose them through a deployed API/dashboard.

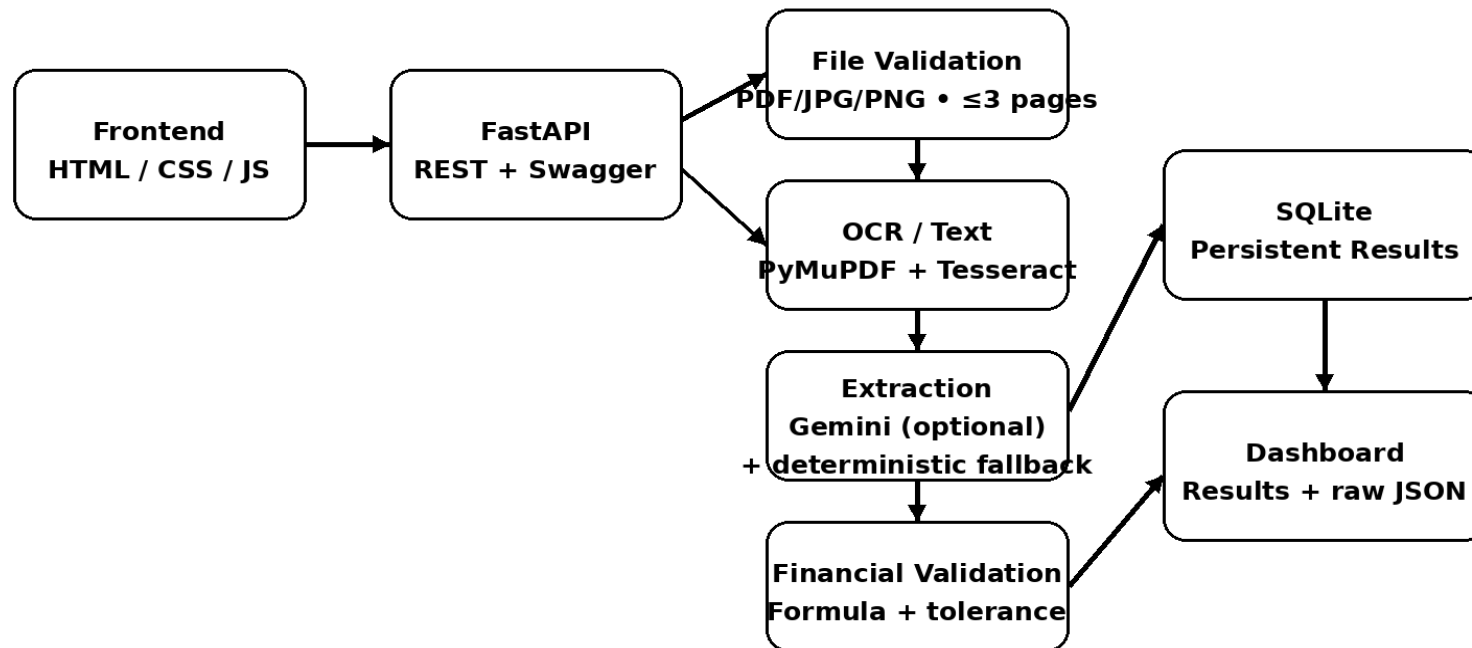
Supported Documents

- Invoice — fields + line items
- Balance Sheet — periods, currency, every visible line item, assets/liabilities/equity
- Profit & Loss — revenue, COGS, gross profit, operating expenses/profit, tax, net profit
- Cash Flow — operating/investing/financing flows, opening/net change/closing cash

Architecture

- Frontend → FastAPI → file validation → OCR/text → AI extraction → financial validation → SQLite → dashboard/API
- Optional Gemini path improves extraction while deterministic parsing provides a keyless fallback.
- Evidence and page numbers are retained where available.

DocIntel — Document Intelligence Architecture



Flow: Upload → Validate → Extract/OCR → Structure → Validate finances → Persist → Dashboard/API

Validation Engine

- Invoice: $\text{subtotal} + \text{tax} - \text{discount} \approx \text{total}$; $\text{line-item sum} \approx \text{subtotal}$.
- Balance Sheet: $\text{liabilities} + \text{equity} \approx \text{assets}$.
- P&L: $\text{revenue} - \text{COGS} \approx \text{gross profit}$; $\text{gross profit} - \text{OPEX} \approx \text{operating profit}$; $\text{operating profit} - \text{tax} \approx \text{net profit}$.
- Cash Flow: $\text{operating} + \text{investing} + \text{financing} \approx \text{net change}$; $\text{opening} + \text{net change} \approx \text{closing}$.
- Missing inputs return NOT_APPLICABLE; tolerance is configurable in code.

API & Persistence

- POST /api/v1/documents/process — multipart upload + document_type
- GET /api/v1/documents/{document_name} — latest stored result
- GET /api/v1/documents — dashboard list
- GET /api/v1/health — health check
- Swagger/OpenAPI available at /docs. SQLite stores structured JSON and metadata.

Frontend

- Professional responsive dashboard with document type selection, upload/process action and API health.
- Processed-document table, status indicators, extracted fields, evidence, line items, validation table and raw JSON.
- Low/missing fields and validation failures are clearly surfaced.

Testing, Security & Reliability

- Automated tests cover validation, extraction/financial calculation and API health.
- Controlled errors prevent stack traces from reaching users.
- No secrets in source; environment variables for model keys.
- Upload type, readability, page count and size are checked before extraction.

Deployment & Production Improvements

- Docker + Render configuration included.
- For stronger production persistence, replace SQLite with PostgreSQL.
- Future: async jobs, authentication/rate limits, managed Document AI, object storage, audit trails, tracing and evaluation metrics.

AI Usage & Interview Readiness

- The assessment permits ChatGPT, Claude, Gemini, Copilot, Cursor and similar tools.
- Candidate should be able to explain architecture, prompts/model choices, OCR, validation, API design and deployment.
- Documented AI/tool usage should be updated honestly before submission.